Decentralized Algorithm for Joint Data Placement and Rate Allocation in Content-Aware Networks

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Abstract. In this paper a mathematical model of data delivery in Content-Aware Network (CAN) is studied. Such network architecture assumes that users access data objects (content) replicated on a number of cache servers. A CAN operator decides on the placement of data objects on behalf of the content publishers, and decides on the transmission rate allocation of users (content consumers downloading content, and publishers uploading and updating their content). A problem of minimizing the mean transmission delay is considered and formulated as a mixed-integer programming problem. Decomposition-based heuristic is proposed, which utilizes projected gradient descent and greedy strategy. The algorithm is presented as a decentralized protocol, embedded in the network architecture.

Keywords: Content-Aware Network, flow control, facility location.

1 Introduction

The problem of optimal planning of data delivery networks in the Internet has been studied systematically in the last decade, in response to rapidly increasing popularity of the Web [1]. The various formulations of placement problems of servers, proxies and caches in computer networks have been studied in e.g.: [2,3]. The first Content Delivery Network was deployed by Akamai Technologies, as a result of the pioneering research in this area [4]. Since then, most of the major content providers entrusted their offered services to such systems. Recently, several new system architectures for future Internet have been proposed, such as Content Centric Network [5] and Content-Aware Networks, optimized for delivery of large volumes of content [6].

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In this paper a mathematical model of Content-Aware Network (CAN) is presented. In such network the transit routers are supplied with special cache servers, which store the content replicas generated on behalf of the content publishers. The performance of the considered system is expressed as a mean transmission delay between users and servers. A decomposition-based heuristic algorithm is presented, which allows for sequential solving the rate allocation [7,8] and content placement [9,10] problems, utilizing projected gradient descent and greedy strategy, respectively.

2 Problem Statement

In the considered problem given is a set of users (called *clients*) who want to read data objects and a set of users (called *publishers*) who want to have their data content delivered to the clients. Clients never read data directly from the publishers' servers. Instead, publishers place the data objects inside public cache servers, connected to the network transit routers. These cache servers are under control of network *operator*, who manages the placement of content on behalf of publishers, as well as assigns transmission rates. Further in this paper the term *user* refers to the local area network (LAN) which generates a packet *flow*, and can be either client user LAN or publisher user LAN.

The description above illustrates the concept of data delivery in *Content-Aware Networks*. The following decisions are to be made by the network operator:

1. how to allocate transmission rates to the users (clients and publishers),
2. in which cache servers to place individual data objects,
3. how to redirect clients to the cache servers.

All those decisions influence the system's performance in terms of request delays perceived by users (clients and publishers). It is assumed that the objective of network operator's decisions is to minimize the sum of mean delays. The formulation is presented in the next subsections and the summary of notation is contained in Table 1.

2.1 Objective

The total delay, associated with mth user's requests concerning nth object on server s, is denoted by $\tau_{mns}(\mathbf{x}, \mathbf{y})$ and it can be given with the formula:

$$\tau_{mns}(\mathbf{x}, \mathbf{y}) = \tau^{(1)}_{mns}(y_{mns}) + \sum_{l=1}^{L} a_{mnsl} \left(\tau^{(2)}_{l}(\mathbf{x}, \mathbf{y}) + \tau^{(3)}_{l} \right)$$

where $\tau^{(1)}_{mns}(y_{mns})$ is a time it takes to push all nth object's bits onto link by the source node, $\tau^{(2)}_{l}(\mathbf{x}, \mathbf{y})$ is a queue sojourn delay (incorporating packet's waiting and serving time on the router's interface), $\tau^{(3)}_{l}$ are the rest of constant delays

Table 1. Summary of notation

Symbol	Description
$m \in \{0, 1, \ldots, M\}$	user's index, where $m = 0$ denotes publisher and $m \in \{1, \ldots M\}$ denotes client,
$n \in \{1, \ldots, N\}$	data object's index,
$s \in \{1, \ldots, S\}$	cache server's index,
$l \in \{1, \ldots, L\}$	transmission link's index,
$x_{mns} \in \{0, 1\}$	assignment of m-th client to s-th server for reading object n,
$x_{0ns} \in \{0, 1\}$	assignment of n-th object the cache server s $x_{ns} = [x_{0ns}]$, $n = 1, 2, \ldots, N, s = 1, 2, \ldots, S$, $\mathbf{x} = [x_{mns}]$, $m = 0, 1, \ldots, M, n = 1, 2, \ldots, N, s = 1, 2, \ldots, S$,
$y_{mns} \geq 0$	rate allocated to the m-th client for downloading n-th object from server s,
$y_{0ns} \geq 0$	rate allocated to the publisher of n-th object for uploading this object to server s, $\mathbf{y} = [y_{mns}]$, $m = 0, 1, \ldots, M, n = 1, 2, \ldots, N, s = 1, 2, \ldots, S$,
$y_{mns,\max} \geq 0$	maximum achievable rate for transmitting n-th object from/to m-th user to/from s-th server; here it is assumed to equal: $y_{mns,\max} = \min_{l \in \{l: a_{mnsl}=1\}} C_l$,
$a_{mnsl} \in \{0, 1\}$	routing variable, which defines if user m uses link l while downloading (uploading) object n from (to) server s,
C_l	capacity of physical link l,
b_n	size of object n

related to the link l (e.g. processing delay, propagation delay). So, the objective function is given as:

$$Q(\mathbf{x}, \mathbf{y}) = \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} x_{mns} \tau_{mns}(\mathbf{x}, \mathbf{y}).$$

2.2 Constraints

The following requirements imposed by the problem may be formulated:

1. Each user must be able to download every object from one location:

$$\forall_{m \in \{1,\ldots,M\}} \forall_{n=1,\ldots,N} \quad \sum_{s=1}^{S} x_{mns} = 1 \ . \tag{1}$$

2. There must be a copy of an object located on the server for which this user is assigned to:

$$\forall_{m \in \{1,\ldots,M\}} \forall_n \forall_s \quad x_{mns} \leq x_{0ns} \ . \tag{2}$$

3. The links' capacities must not be exceeded:

$$\forall_{l=1,\ldots,L} \quad \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} a_{mnsl} x_{mns} y_{mns} \leq C_l \ . \tag{3}$$

Observe that constraints (1) and (2) imply that at least one copy of each object must be located in the network, i.e. $\forall_{n=1,\ldots,N} \quad \sum_{s=1}^{S} x_{0ns} \geq 1$. In this paper, it is assumed, that storage capacity on every server is big enough to keep all the objects (i.e. $\forall_{s\in\{1,\ldots,S\}} \sum_{n=1}^{N} b_n \leq B_s$, where B_s is the storage capacity of server s). Without such assumption the problem becomes much more difficult and hard to approximate [11], however it appears that mass storage is the least deficient resource in practice.

2.3 Problem Formulation

The joint object placement and capacity allocation problem in Content-Aware Network is formulated as follows. For given values of $M, N, S, L, \mathbf{A} = [a_{mnsl}]$, $\mathbf{C} = [C_l]$ one should find:

$$(\mathbf{x}^*, \mathbf{y}^*) = \arg\min_{\mathbf{x},\mathbf{y}} Q(\mathbf{x}, \mathbf{y})$$

such that constraints (1)–(3) hold.

This is nonlinear mixed integer programming problem (discrete-continuous). Even assuming that $\mathbf{y}^*$ is known in advance the problem is NP-hard with respect of $\mathbf{x}$. To obtain exact solution, algorithms such as branch-and-bound or outer approximation [12] may be applied. However, such approach leads to solving procedures with prohibitive running time (requiring exponential number of steps in terms of M, N, S and L). Moreover, these procedures are hard to implement effectively in decentralized manner. In this paper, a particular heuristic approach based on problem decomposition is proposed and its decentralized version is presented. The general formulation of objective functions allows for plugging functions representing the time spent by packets inside routers' queues arbitrarily. However, in this paper it is assumed that all routers adhere to M/M/1 queueing characteristic, i.e. mean sojourn time of packet in link's l router is $\tau_l^{(2)}(\mathbf{x}, \mathbf{y}) \approx (C_l - \sum_{m,n,s} x_{mns} y_{mns})^{-1}$. This is justified for FTP-type transactions, and it is assumed that such traffic dominates the transmissions in Content-Aware Networks. Thus solving the problem formulated in this paper can be seen as optimizing the network with respect to the static file transfer applications. For other types of traffic different queueing models need to be taken into consideration [13].

3 Solution Algorithms

An additional specificity of the problem is the decentralized architecture of the system. It is required that the solution algorithms can be distributed across the network devices, since both input data, as well as decision variables, are bound to spatially distributed network nodes. The implementation should assume the limited communication between entities cooperating in the computation of the global solution. In this section the idea of decomposition is presented, followed by the description of two-level solution algorithm, and its implementation in the decentralized model. Such implementation consists of routines which must be implemented on specified network nodes, communicating via fixed protocol.

3.1 Two-Level Heuristic Algorithm

Given the complexity of the problem, efficient approximate solution algorithms are desirable. An observation that the discrete subproblem (i.e. finding $\mathbf{x}$) and continuous subproblem (i.e. finding $\mathbf{y}$) alone are tractable motivates the following heuristic approach. First, the problem is decomposed into two subproblems:

1. (upper level problem) The rate allocation problem which is solved for a fixed content location, i.e. for given $M, N, S, L, \mathbf{A} = [a_{mnsl}], \mathbf{C} = [C_l], \mathbf{x}'$, find:

$$\mathbf{y}' = \arg\min \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} x'_{mns} \tau_{mns}(\mathbf{x}', \mathbf{y})$$

   subject to constraint (3) (assuming $\mathbf{x} = \mathbf{x}'$).
2. (lower level problem) The data placement problem which is solved for a particular capacity allocation in virtual networks, i.e. for given $M, N, S, \overline{\tau} = [\overline{\tau}_{mns}]$, find:

$$\mathbf{x}' = \arg\min \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} x_{mns} \overline{\tau}_{mns}$$

   subject to constraints (1)–(2), where $\overline{\tau}_{mns}$ is equal to $\tau_{mns}(\mathbf{x}, \mathbf{y})$, where $\mathbf{y}$ is calculated as:

$$y_{mns} = \begin{cases} y'_{mns} & \text{if } x_{mns} = 1, \\ y_{mns,\max} & \text{if } x_{mns} = 0, \end{cases}$$

   and $\mathbf{x}$ being the previous value of $\mathbf{x}$, i.e. those for which $\mathbf{y}'$ was obtained.

The data placement in cache servers influences the need for capacity in different links. At the same time, any particular allocation of transmission rates causes different load characteristics in the network and on cache servers. Since the solution of one problem gives the input data for another one, procedures for solving these both subproblems may be repeated alternately, as long as the value of objective function is decreasing. The upper and lower level problems may be solved with any known solution algorithm.

However, even if both problems are solved to optimality, it is not guaranteed in general that such approach will result in optimal solution to the global problem. Moreover, one should notice that even if the solution of the lower level was given with dependence on $\mathbf{x}$, the lower level would not provide the optimal solution of the problem. That is because the objective function for the lower level problem is linearized by assuming constant value of τ_{mns} (while it depends on $\mathbf{x}$). Nevertheless, such approach allows for elegant decomposition and decentralization of the solution, supporting practical implementation.

In this paper it is proposed to use greedy approximation algorithm on the lower level and asymptotically exact algorithm on the upper level. First, the centralized approach is presented, which serves as a basis of the decentralized version of the algorithms.

3.2 Centralized Algorithm Based on Two-Level Heuristic

Rate Allocation Algorithm. The presented algorithm for the rate allocation subproblem requires that the delay functions $\tau_{mns}(\mathbf{x}', \mathbf{y})$ are convex with respect to $\mathbf{y}$. Assume that $\mathbf{x}$ is fixed and equal to $\mathbf{x}'$. Let us define Lagrange function:

$$L_1(\mathbf{y}, \lambda) = \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} x'_{mns} \tau_{mns}(\mathbf{x}', \mathbf{y}) +$$

$$+ \sum_{l=1}^{L} \lambda_l \left(\sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} a_{mnsl} y_{mns} x_{mns} - C_l \right)$$

where $\lambda = [\lambda_1, \ldots \lambda_L]^T$ is the vector of Lagrange multipliers.

The Langrange function's derivatives are as follows:

$$\frac{\partial L_1(\mathbf{y}, \lambda)}{\partial y_{mns}} = \sum_{i=0}^{M} \sum_{j=1}^{N} \sum_{k=1}^{S} \left(x'_{ijk} \frac{\partial \tau_{ijk}(\mathbf{x}', \mathbf{y})}{\partial y_{mns}} \right) + x'_{mns} \sum_{l=1}^{L} a_{mnsl} \lambda_l \tag{4}$$

$$\frac{\partial L_1(\mathbf{y}, \lambda)}{\partial \lambda_l} = \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} a_{mnsl} x'_{mns} y_{mns} - C_l \ . \tag{5}$$

Equating above derivatives to 0 and solving the set of equations with respect to y_{mns} and λ_l one can obtain the optimal solution of lower level problem (from KKT conditions [14], the only points where derivatives of convex function are zero coincide with the global optimum). But generally, due to nonlinearity, it is hardly possible to obtain analytical solution of such system. Consequently, the projected gradient method is proposed to be applied [15].

The following iterative solution algorithm is proposed. For each variable y_{mns} and λ_l a sequence of stepwise updates are applied, starting from any feasible point, according to the following formulas:

$$y_{mns}[t+1] = y_{mns}[t] - K_{ymns} \left(\sum_{i=0}^{M} \sum_{j=1}^{N} \sum_{k=1}^{S} \left(x'_{ijk} \left. \frac{\partial \tau_{ijk}(\mathbf{x}', \mathbf{y})}{\partial y_{mns}} \right|_{(\mathbf{y}, \lambda) = (\mathbf{y}[t], \lambda[t])} \right) + \right. \tag{6}$$

$$\left. + x'_{mns} \sum_{l=1}^{L} a_{mnsl} \lambda_l[t] \right)$$

$$\lambda_l[t+1] = \lambda_l[t] - K_{\lambda l} \left(\sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} a_{mnsl} x'_{mns} y_{mns}[t] - C_l \right) \tag{7}$$

where K_{ymns} and $K_{\lambda l}$ are parameters controlling the rate of convergence (for sufficiently small values it is guaranteed that algorithms will converge, however may require a large number of steps).

It is clear that computation of (7) can be executed locally on each link (or rather on an appropriate router). Thus the decentralization of the whole algorithm relies on the possibilities of distributing iterative routine (6). This inherently depends on the particular form of the delay functions $\tau_{mns}(\mathbf{x}, \mathbf{y})$.

Data Placement Algorithm. Recall that in our considerations all cache server storages are large enough to hold any number of objects. Assuming fixed $\mathbf{y}$, equal to $\mathbf{y}'$, the problem becomes well-studied *uncapacitated facility location problem* [16]. In this subsection it is presented how to efficiently obtain a good solution using centralized algorithm. The approach is based on 2-approximation algorithm given in [17].

Relaxing the integrality constraints on $\mathbf{x}$, the Lagrange function with respect to the remaining constraints (1)–(2) takes form:

$$L_2(\mathbf{x}, \phi, \psi) = \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} x_{mns} \overline{\tau}_{mns} + \sum_{m=0}^{M} \sum_{n=1}^{N} \phi_{mn} \left(1 - \sum_{s=1}^{S} x_{mns} \right) + \\ + \sum_{m=1}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} \psi_{mns} (x_{0ns} - x_{mns})$$

where $\overline{\tau}_{mns}$ is equal to $\tau_{mns}(\mathbf{x}, \mathbf{y})$ for given $\mathbf{y}$ (a solution of the lower level problem) and previous value of $\mathbf{x}$. Using the Lagrange multipliers as decision variables the following dual problem can be obtained:

$$\text{maximize} \quad \sum_{m=1}^{M} \sum_{n=1}^{N} \phi_{mn} \tag{8}$$

subject to:

$$\forall_{m=1,\ldots,M} \; \forall_{n=1,\ldots,N} \; \forall_{s=1,\ldots,S} \quad \phi_{mn} - \psi_{mns} \leq \overline{\tau}_{mns} \;\; ,$$

$$\forall_{n=1,\ldots,N} \; \forall_{s=1,\ldots,S} \quad \sum_{m=1}^{M} \psi_{mns} \leq \overline{\tau}_{0ns} \;\; ,$$

$$\forall_{m=1,\ldots,M} \; \forall_{n=1,\ldots,N} \; \forall_{s=1,\ldots,S} \quad \phi_{mn} \geq 0, \psi_{mns} \geq 0 \;\; .$$

Applying greedy strategy to determine the solution of the dual problem allows for finding $\mathbf{x}$ satisfying complementary slackness conditions of the relaxed problem. Such solution can be shown to be close to the optimal of the original (discrete) problem (assuming metric weights in the input data). The algorithm is described in detail in Sect. 3.3.

Initialization. The application of two presented stages of heuristic requires an initial feasible solution to start with. Before the transmission rates can be computed, it is required to give a feasible data placement and user assignment. This can be accomplished by running the lower level algorithm first, assuming no traffic in the network. This corresponds to solving the data placement problem for the input data consisting of "best-case" delays, that is ones obtained by assuming all maximal rates and zero queuing delays (only constant link propagation delays). Thus initialization input data becomes:

$$\tau_{mns} = \frac{1}{y_{mns,\max}} + \sum_{l} a_{mnsl} \tau_l^{(3)} \;\; .$$

For the initial $\mathbf{x}$, the heuristic continues by solving upper level subproblem.

3.3 Decentralized Algorithm for Special Case

Decentralized Rate Allocation Algorithm. Let α_{mns} defines fraction of object sent in one packet (i.e. $\alpha_{mns}b_n$ is the total number of packets to be sent). Then delay of file transfer, corresponding to the transmission with rate y_{mns}, is given by:

$$\tau^{(1)}_{mns}(y_{mns}) = \frac{\alpha_{mns}b_n}{y_{mns}} \quad . \tag{9}$$

Assuming that all queues in routers are M/M/1, the queuing delay may be expressed with the following formula [18]:

$$\tau^{(2)}_{l}(\mathbf{x},\mathbf{y}) = \frac{1}{C_l - \sum_{m=0}^{M}\sum_{n=1}^{N}\sum_{s=1}^{S} a_{mnsl}x_{mns}y_{mns}} \quad . \tag{10}$$

Consequently, the total delay, associated with mth user's requests concerning nth object on server s is given by the formula:

$$\tau_{mns}(\mathbf{x},\mathbf{y}) = \frac{\alpha_{mns}b_n}{y_{mns}} + \sum_{l=1}^{L} a_{mnsl}\left(\frac{1}{C_l - \sum_{i=0}^{M}\sum_{j=1}^{N}\sum_{k=1}^{S} a_{ijkl}x_{ijk}y_{ijk}} + \tau^{(3)}_{l}\right) . \tag{11}$$

Since the function $f(z) = \frac{1}{z}$ is convex for $z > 0$, thus (9) is also convex. Moreover, taking into account the convexity preservation of affine mappings, (10) is also convex. Finally, the objective function (11) is also convex since it is linear combination (with positive coefficients) of convex functions [14].

Now the derivative (5) may be calculated, and the algorithm (6) may be rewritten as follows:

$$y_{mns}[t+1] = y_{mns}[t] - K_{ymns}\left(x'_{mns}\left(\frac{\alpha_{mns}b_n}{(y_{mns}[t])^2} + \right.\right. \tag{12}$$

$$\left.\left. + \sum_{i=0}^{M}\sum_{j=1}^{N}\sum_{k=1}^{S}\sum_{l=1}^{L}\frac{a_{ijkl}a_{mnsl}x'_{mns}}{\left(C_l - \sum_{e=0}^{M}\sum_{f=1}^{N}\sum_{g=1}^{S} a_{efgl}x'_{efgs}y_{efg}[t]\right)^2}\right)\right) +$$

$$\left. + x'_{mns}\sum_{l=1}^{L} a_{mnsl}\lambda_l[t]\right) \quad .$$

The formula (12) may be rewritten using (9) and (10):

$$y_{mns}[t+1] = y_{mns}[t] - K_{ymns}\left(x'_{mns}\left(\frac{(\tau^{(1)}_{mns}(y_{mns}))^2}{\alpha_{mns}b_n} + \right.\right. \tag{13}$$

$$\left.\left. + \sum_{i=0}^{M}\sum_{j=1}^{N}\sum_{k=1}^{S}\sum_{l=1}^{L} a_{ijkl}a_{mnsl}x'_{mns}(\tau^{(2)}_{l}(\mathbf{x},\mathbf{y}))^2\right) + x'_{mns}\sum_{l=1}^{L} a_{mnsl}\lambda_l[t]\right) \quad .$$

Each source may easily determine the time $\tau^{(1)}_{mns}(y_{mns})$, but to calculate (13), the values of queuing delays (10) and congestion signals (7) along the route

must be known. The congestion signals may be estimated e.g. basing on the loss probability (when some Active Queue Management procedures are implemented in the network [19]). The delays in the network may be monitored (e.g. using Internet Control Message Protocol Echo Request or other utilities).

Algorithm 1. User algorithm for upper level subproblem

It is assumed that user m knows routing matrix entries a_{mnsl} corresponding to its path to all servers, as well as its own current assignments x'_{mns}. Moreover, values α_{mns}, b_n, K_{ymns} are known.
The procedure is executed by source nodes of every active flow (i.e. mth user request for object n from server s if $x_{mns} = 1$):

1. Let $t = 0$, $\epsilon > 0$.
2. Compute $\tau^{(1)}_{mns}(y_{mns}[t])$ according to (9).
3. Receive $\tau^{(2)}_l$ from all links (routers) on the path to server s.
4. Receive the value $x'_{mns} \sum_l a_{mnsl}\lambda_l[t]$ from the network (routers).
5. Update transmission rate $y_{mns}[t+1]$ according to (13).
6. If $|y_{mns}[t+1] - y_{mns}[t]| < \epsilon$ then STOP.
7. Increase t. Go to 2.

Algorithm 2. Link algorithm for upper level subproblem

The procedure below is executed in link l (routers' software):

1. Let $t = 0$.
2. Measure the total traffic on the link, i.e. $\sum_{m,n,s} a_{mnsl}x'_{mns}y_{mns}[t]$.
3. Update congestion signals $\lambda_l[t+1]$ according to (7).
4. Send congestion signals to all users transmitting data through the link l.
5. Increase t. Go to 2.

Decentralized Data Placement Algorithm. Observe that the linear program (8) is equivalent to N linear programs for each $n = 1, \ldots, N$, thus without the loss of generality one can assume that n is fixed and solve for each n independently. The complementary slackness conditions imply that the solution $\mathbf{x} = [x_{mns}]$ is optimal if and only if:

$$\forall_{m=1,\ldots,M} \forall_{n=1,\ldots,N} \ \forall_{s=1,\ldots,S} \quad x_{mns} = 0 \ \vee \ \phi_{mn} = \psi_{mns} + \overline{\tau}_{mns}$$

$$\forall_{n=1,\ldots,N} \ \forall_{s=1,\ldots,S} \quad x_{0ns} = 0 \ \vee \ \sum_{m=1}^{M} \psi_{mns} = \overline{\tau}_{0ns}$$

which suggests the following procedure. Starting from $\phi = \mathbf{0}$, $\psi = \mathbf{0}$, for each user m increase the dual variable ϕ_{mn} until the "connection cost" to any server s is paid (this cost equals to $\overline{\tau}_{mns}$). After this, increase the dual variable ψ_{mns} for such server s, as long as the sum of contributions $\sum_m \psi_{mns}$ is equal to the "object placement cost", equal to $\overline{\tau}_{0ns}$. This can be translated into following greedy

Algorithm 3. User algorithm for lower level subproblem

The procedure below is executed by every user's node m and for every data object n requested by that user, until all clients cease to be unconnected:

1. (initialization) Let $\delta > 0$, $t = 0$, $\phi_{mn}[0] = 0$, $\psi_{mns}[0] = 0$, $x_{mns} = 0$ for all $s = 1, \ldots, S$.
2. Increase t.
3. Probe the neighbor servers to estimate the current value of $\overline{\tau}_{mns}$.
4. Set $\phi_{mn}[t] \leftarrow \phi_{mn}[t-1] + \delta$.
5. For all servers s:
   if $\phi_{mn}[t] \geq \overline{\tau}_{mns}$ then set $\psi_{mns}[t] \leftarrow \psi_{mns}[t-1] + \delta$ else $\psi_{mns}[t] = 0$.
6. Send the value $\phi_{mn}[t]$ and $\psi_{mns}[t]$ to the server s.
7. Receive messages x_{0ns} from servers. If m is unconnected, and if for some s, $x_{0ns} = 1$, then connect to the server, i.e. put $x_{mns} = 1$, and go to the step 8. Otherwise repeat from the step 2.
8. Increase t. Set $\phi_{mn}[t] \leftarrow \phi_{mn}[t-1]$ and $\psi_{mns}[t] \leftarrow \psi_{mns}[t-1]$.
9. Send the values $\phi_{mn}[t]$ and $\psi_{mns}[t]$ to the server s.
10. Receive messages x_{0ns} from servers. If for some server s', $x_{0ns'} = 1$ and the estimated value $\overline{\tau}_{mns} - \overline{\tau}_{mns'} > 0$, then reconnect to the server s', i.e. put $x_{mns} = 0$ and $x_{mns'} = 1$.
11. Go to step 8.

scheme: for each object n on server s sort all unconnected clients m by their connection costs $\overline{\tau}_{mns}$. Now consider all clusters consisting of a server and a subset of "cheapest" clients, denoting the set of such client indices by D_{ns}. Select the most cost-effective cluster, i.e. one minimizing the ratio $(\sum_{m \in D_{ns} \cup \{0\}} \overline{\tau}_{mns})/|D_{ns}|$. Such server is selected to cache the object and all clients from D_{ns} are assigned to it. The process is repeated for the remaining unconnected clients, with the exception, that all already selected servers have cost $\overline{\tau}_{0ns}$ substituted by 0.

Such algorithm allows to find an object placement x_0 and users' assignment $\mathbf{x}$ to the object replicas, which is usually very close to the optimal solution. Under the assumption that the input data matrix $\overline{\tau}_{mns}$ satisfies the triangle inequality, it can be shown that such algorithm gives the solution no worse than 2 times the optimal value [17]. Decentralized implementation is presented as Algorithm 3 (users' routine) and Algorithm 4 (cache servers' routine).

4 Computational Example

Simulation study was carried out for an example network, consisting of two cache servers, two user LANs and one publisher. The network consists of two routers with all link capacities equal to 10 MB/s. Results of this experiment are presented in the Fig. 1. Although global minimum is not achieved (the final value is very close to optimal), the heuristic approach requires to search in only a fraction of feasible solution space to end up with satisfactory outcome. It can be seen that subsequent iterations gradually improve the solution. This example supports the legitimacy of presented idea.

Algorithm 4. Cache server algorithm for lower level subproblem

The procedure below is executed by every cache server's node s and for every data object n:

1. (initialization) Let $t = 0$, $x_{0ns} = 0$, $\psi_{mns}[0] = 0$.
2. Receive messages $\phi_{mn}[t]$ and $\psi_{mns}[t]$ from all clients.
3. If $\sum_m \phi_{mn}[t] = \sum_m \phi_{mn}[t-1]$ then STOP.
4. If $\sum_m \psi_{mns}[t] \geq \overline{\tau}_{0ns}$ then cache object, i.e. put $x_{0ns} = 1$, and send this information to all such clients m that $\psi_{mns}[t] > 0$.
5. Increase t. Go to 2.

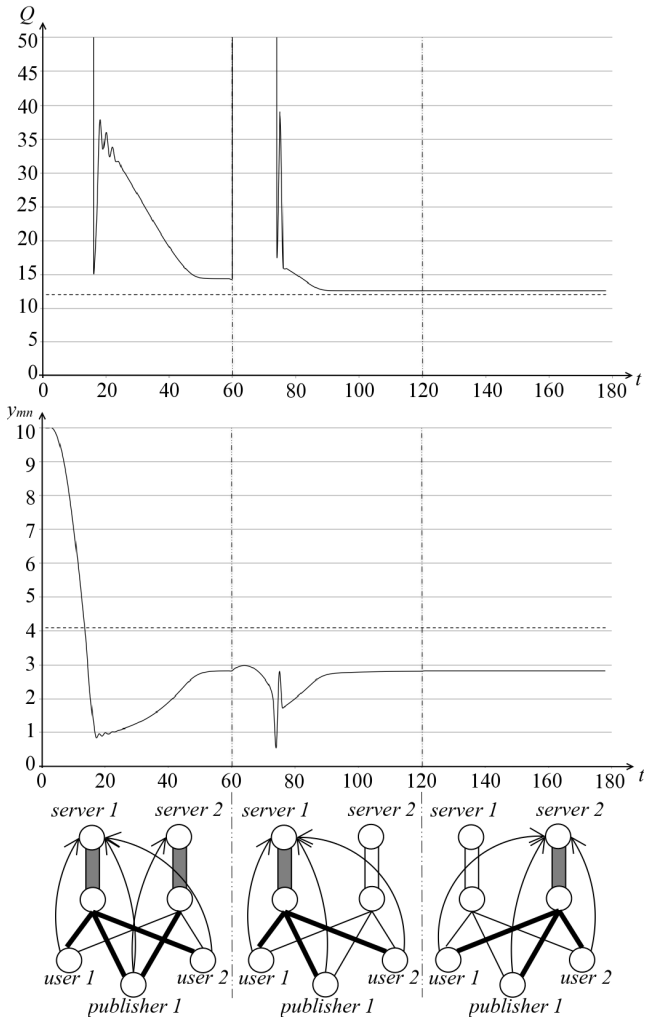

Fig. 1. Example: 3 iterations of heuristic algorithm (iterations are separated by dotted vertical lines). The value of objective function Q (topmost plot) can be seen to converge very closely to optimal value (horizontal dashed line). The process of control of transmission rate $y_{mn} = \sum_s x_{mns} y_{mns}$ is presented in the middle plot. Object assignment configurations in all iterations are shown in the bottom plot.

5 Conclusions

Content delivery networks provide a number of challenges in the area of computer network optimization. In order to minimize the mean delay of transmitting data between user participants (client and publisher LANs) and caching servers, CAN operators need to efficiently approximate solutions to hard mixed-integer programming problems. Such task can be accomplished in the real-time by applying decentralized protocols, which incorporate adequate optimization algorithms. One such solution, based on two-level decomposition is given in this paper.

The presented algorithm optimizes the rate allocation policy with respect to transfer of static content. In case of small sized files the queuing delays on routers contribute a significant fraction of total transfer time, thus the optimal solutions tend to avoid assigning many flows to the same links. In case of large files the queuing delays are often negligible, and such flows require allocating links with large capacity in order to reduce the total transfer time.

Future works will include the incorporation of market-based and utility-based analysis to the model of CAN, in order to establish the pricing strategies for operators. Additionally, server storage capacities need to be taken into consideration, as the model presented in this paper simplifies the operational properties of cache servers.

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Development of Service Composition by Applying ICT Service Mapping

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Abstract. In this paper, the problem of ICT service mapping in service composition process is highlighted. In many cases, especially for telecommunication operators, it is important to allow service providers to compose services which support data transfer and processing. In this work a general service composition process is outlined and the ICT service mapping task is described. Next, the solution is proposed. At the end the case study is presented and conclusions are drawn. The main contribution of this paper is the proposition of decision tables as a tool for ICT service mapping.

Keywords: service mapping, service composition, communication service, computational service.

1 Introduction

Recently, enterprises tend to collaborate and integrate their business cores in Web markets in order to maximize both clients' satisfaction and own profits [1]. Therefore, there is an increasing need to develop methods for combining existing services[1] together to enrich functionalities and decrease execution costs [2]. Hence old-fashioned way of developing composite services, i.e., human-based service designing, becomes very insufficient due to enormous number of available services and lack of computational resources. That is why the automatic or semi-automatic *service composition* method is a crucial element of any system that implements service-oriented architecture (SOA) [1,2].

In general, service composition process consists of two steps [4]. First, the required functionalities and their interactions, i.e., control and data flow, are identified. Second, the execution plan is established, i.e., for sets of functionalities an appropriate version of service is selected. The service selection is accomplished basing on non-functional properties which in general can be referred to as *Quality-of-Service* (QoS) attributes.

[1] *Services* are self-describing, stateless, modular applications that are distributed across the Web [1,2]. The services provide functionalities and are described by quality attributes called *Quality-of-Service* (QoS), e.g., price, reputation, reliability, response time [1,3]. A service that provide single functionality is called *atomic service*, and a composition of atomic services is called a *composite services*.

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